

Session 5 — Live demos: cleaning FLUID-ICU and building Table 1 in R

Foundations of Clinical Research · Weekend 3, Saturday · Neudata · instructor script

	Part A — cleaning	Part B — Table 1 and figures
Slot	5.2 <i>Meet the course dataset</i> (about 0:55 on the clock)	5.4 <i>Describe the course dataset</i> (about 2:02)
Duration	8 minutes	6 minutes
Script	Course/Data/R/01_clean.R	Course/Data/R/02_describe.R
Participants	watch and predict; they do not run R	same

> **FLUID-ICU is SIMULATED.** Say so every time you open the data: no real patient, hospital or country. All numbers below are from `Course/Data/key_results.md` and `Course/Data/cleaning_log.md`.

Before the session (night before + 10 minutes before)

- [] R (4.x) and RStudio installed on the projector laptop; packages: tidyverse, janitor, gtsummary, flextable, survival, survminer, mice, broom.
- [] Open `Course/Data/FLUID-ICU.Rproj` (sets the working folder). Run `01_clean.R` and `02_describe.R` once end to end; both must finish without error.
- [] Increase the editor and console font (Tools → Global Options → Appearance → 18–20 pt) so the back row can read.
- [] Keep the fallback figures open in an image viewer: `Course/Resources/figures/en/data/` (`raw_vs_clean.png`, `table1_snapshot.png`, `fluid_balance_hist.png`, `sofa_by_group.png`, `los_skewed.png`) and `Course/Data/output/figures/`.
- [] Open `Course/Data/cleaning_log.md` and `data_dictionary.md` in a browser tab: you will show them.

Rules of the demo: run one block at a time (Ctrl+Enter); before each block, ask the room "what do you expect?"; after it, say *what was fixed and that it goes in the log*. Never type new code live — the scripts are complete.

Part A — From raw to clean (01_clean.R, 8 minutes)

Step (script block)	What you show	Expected output	Say
0. Read raw as text	<code>dim(raw)</code> , <code>glimpse(raw)</code>	623 rows , 29 columns, everything <chr>	"The study enrolled 620 patients. Why 623 rows?"
1. Column names	<code>names(dat)</code>	short lower-case names (<code>map_mmhg</code> , <code>vent_free_days_28</code>)	"Names without spaces: same in every script."
2. Duplicates	<code>get_dupes(study_id)</code>	3 study IDs appear twice, identical rows; after <code>distinct()</code> 620	"Each patient counted once."

Step (script block)	What you show	Expected output	Say
3. Sites	<code>count(site)</code> before/after	variants such as <code>icu a</code> , ICU-A → 4 sites	"Otherwise R sees many different ICUs."
4. Dates	the three formats	2023-03-14, 14/03/2023, 14-Mar-2023 → one date; 0 failures	"Is 03/04 March or April? Never guess."
5-6. Sex and yes/no	<code>tabyl()</code> before/after	many spellings (M, m, male, F, f, "Female"; Y, 1, yes) → Male/Female, Yes/No	"1 and Yes must mean the same thing."
7-8. Numbers and codes	<code>summary()</code> of lactate, MAP, APACHE II	maxima of 999 , minima of -99 → set to NA	"A lactate of 999 would wreck the mean."
9. Units	lactate median by ICU	ICU C about 9 times higher (mg/dL) → divided by 9.01	"Found only because we summarised by ICU. Ask the site lab."
10. Impossible values	range checks	age 210, weight 7 kg, a negative fluid output → NA + query; patient kept	"We cannot guess 21 or 71. We never delete the patient."
12. Derived variables	the fluid-balance formula	<code>cum_fluid_balance_ d3_ml_kg, positive_ balance_d3, aki_any computed</code>	"Derived values are calculated, never typed."
13. What is left	<code>colSums(is.na(dat))</code>	lactate 93 missing; a handful elsewhere	"These are REAL gaps; we handle them in Session 6."
14. Reproducibility	<code>final cat()</code>	Cleaned data identical to the reference clean file: TRUE/ byte-for-byte identical: TRUE	"Anyone can re-run this and get exactly the same data."

If short of time: run Steps 0, 2, 9, 10 and 14 only (4 minutes) and show the cleaning log for the rest.

Close Part A by opening `cleaning_log.md`: *"Every block you saw is one row here: problem, how many, what we did, why."* Then show the slide *From 623 rows to 527 complete cases* (620 enrolled → 618 with a day-3 balance → 527 complete cases).

Part B — Table 1 and figures (02_describe.R, 6 minutes)

Block	What you show	Expected output (key_results.md)	Say
1. Load	<code>dim(icu), count(pos itive_balance_d3)</code>	620 patients; positive balance 373/618 (60.4%) ; 2 without a balance	"618, not 620: two patients have no computable balance."
2. Shape first	mean/SD of age, median/IQR of lactate; quick histograms	age mean 62.6 (SD 15.2) ; lactate skewed	"Symmetric → mean (SD); skewed → median (IQR)."

Block	What you show	Expected output (key_results.md)	Say
3. Table 1	the gtsummary table by group	negative n = 245, positive n = 373; SOFA 6 (4 to 8) vs 8 (6 to 10) ; vasopressor 24.9% vs 55.0% ; ventilation 39.6% vs 57.9% ; lactate 2.2 vs 2.9 mmol/L; heart failure 22.0% vs 11.5% ; missing counts shown; no p-values	"Were the groups alike at the start? No: the positive group was sicker."
4. Histogram	fluid balance by group	median 8.8 ml/kg (IQR -10.3 to 26.2)	"Continuous exposure, cut at 0."
5. Boxplot	SOFA by group, with dots	medians 6 vs 8	"Show every patient."
6. Bar chart	infection source, sorted	respiratory most common	"Categories → sorted bars, not a 3-D pie."
7. Outcomes	described, not compared	28-day mortality 179/620 (28.9%) ; ICU stay mean 7.3 vs median 6.0 days	"Described today, compared tomorrow."

Close Part B: "This Table 1 is in every group's results pack. Your worksheet now: read it as a reviewer would." → Exercise 6.

Fallback plan

Problem	Do this
R or a package fails	Show <code>raw_vs_clean.png</code> and talk through the problems (Part A); show <code>table1_snapshot.png</code> , <code>fluid_balance_hist.png</code> , <code>sofa_by_group.png</code> (Part B). Teaching points are identical.
Projector cannot show RStudio legibly	Use the PNG figures and <code>cleaning_log.md</code> in a browser at 150% zoom.
Running late	Part A: Steps 0, 2, 9, 10, 14. Part B: blocks 3 and 5 only.
A participant asks to run it themselves	Point to <code>Course/Data/README.md</code> and the free <i>Clinical Data Analysis in R</i> course; offer to help after the session.

Anticipated questions

"Is this real data?" No — simulated, designed to behave like ICU data and to show the classic problems. Any manuscript must say so.

"Why not fix the errors in Excel?" Hand edits leave no trace and cannot be repeated; one wrong paste is invisible. A script plus a log is checkable by a reviewer and repeatable by anyone.

"Why set age 210 to missing instead of guessing 21?" It could be 21, 71 or 10; guessing invents data. Set to missing, query the site, correct only with the source document.

"Why no p-values in Table 1?" Table 1 describes the groups. The SAP says so explicitly; in an RCT any difference is chance, in a cohort it is confounding to be adjusted for, not tested.